

Artificial Intelligence Lab (AI1002)

Date: March 19th 2025

Course Instructor(s)

Ms Sarah , Mr Shafique, Mr. Muhammad Khalid,
Mr Abdullah Yaqoob

Lab Mid Exam A

Total Time: 90 minutes

Total Points: 50

Total Questions: 03

Semester: Spring 2025

Campus: Karachi

Department: CS, AI, SE, CY &DS

Instructions:

- Submit a .py file or .ipynb file for all questions.
- Each file must contain a header with your roll number.
- Also submit a ".pdf" OR ".docx" file containing screenshots of codes and clearly show the output.
- All your codes must produce output.
- For Question 03, the code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Student Name

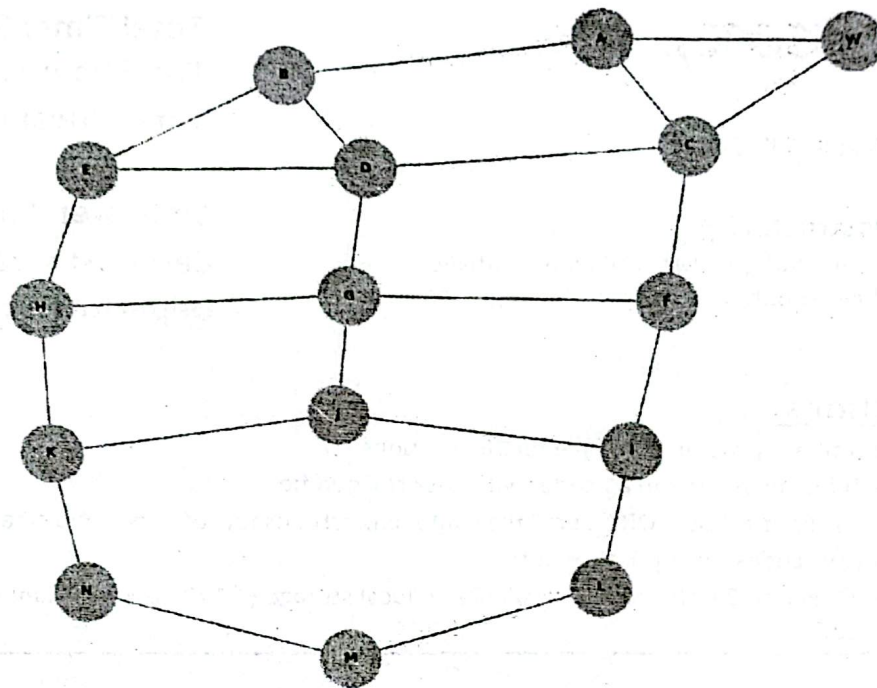
Roll No

Section

Student Signature

LLO # 1: Explain and implement BFS and DFS algorithms, comparing their efficiency and applications in uninformed search. [18 points]

Q1: You are part of a logistics company that employs goal-based agents for efficient route planning in a city. The city map is represented as an undirected graph, where each node represents a delivery location, and the edges represent possible routes between those locations. The company's goal is to design a goal-based agent that finds the path between a starting location; warehouse (W) and the destination (N). You need to use Breadth-First Search (BFS) to solve this.



LLO #3: Apply and implement A* and GBFS algorithms, evaluating their heuristics, efficiency, and applications in informed search. [18 points]

Q2. A robotic agent is deployed inside a structured environment represented as a 6x6 grid-based maze. The robot's goal is to navigate from a given **starting position (0,0)** to a designated **exit point (4,3)** while minimizing the total distance traveled using Manhattan distance. However, the maze contains **obstacles** (represented as 1s in the grid), meaning the robot cannot pass through those cells. The robot can move **up, down, left, or right** but cannot move diagonally. The pathfinding must be efficient, ensuring the shortest possible route is taken using the *Greedy Best first search*

The grid look like :

```
grid = [
    [0, 0, 0, 0, 0, 0],
    [0, 1, 1, 1, 0, 0],
    [0, 0, 0, 1, 0, 0],
    [0, 1, 0, 1, 0, 0],
    [0, 1, 0, 0, 0, 0],
]
```

LLO #4: Implement Genetic Algorithms to solve optimization problems, analyzing their operators, convergence, and performance. [14 points]

Q3.

Instructions:

The code is available on local storage (\\172.16.5.43\Exam Material\ (Your Teacher Name)).

Problem Statement:

In a computer network, several data packets need to be routed through a series of routers to reach their destination. However, due to network congestion, some routers may experience higher loads than others, resulting in delays and inefficiencies. Your task is to implement a **Genetic Algorithm (GA)** to optimize the routing of data packets, minimizing the total network congestion and ensuring efficient data delivery.

You are given a network of routers and a set of possible routes between them. Each route has a bandwidth and a congestion level associated with it. The objective is to find the optimal routing paths for a set of packets such that the total network load (measured by the congestion levels of the routes) is minimized.

The tasks are to be completed with `pop_size = 10` and for 50 generations.

Task:

1. By using Roulette Selection method (individuals are selected for reproduction based on their fitness, with higher fitness individuals having a higher probability of being chosen), create a selection function. [4]
2. You are expected to write a crossover function using Single Point crossover. [4]
3. You are expected to write a mutate function using Binary Mutation (a randomly selected bit in an individual's binary representation is flipped to introduce variation). [4]
4. Update such that the code runs completely and outputs best solution. [2]